Predicting "Do Not Drive" Orders Using Vehicle Recall Data

Ben An, Dan Mosu, Imani Jacobs, Kai Clancy

Part 1: Context and Business Problem

The automotive industry represents one of the largest sectors of the U.S. economy, producing the primary mode of personal transportation for hundreds of millions of consumers. Given the mechanical complexity and scale of modern vehicle manufacturing, latent engineering and assembly defects periodically bypass quality-control gates and reach active consumer fleets. When these safety risks are identified, the National Highway Traffic Safety Administration (NHTSA) coordinates manufacturer-issued vehicle safety recalls.

The vast majority of recalls permit drivers to operate their vehicles with ordinary precautions while waiting for repair parts or scheduling dealership service. In rare, acute instances, however, a defect presents such immediate, catastrophic danger that it triggers a **"Do Not Drive" (DND)** advisory. Unlike standard recalls, a DND order advises or prohibits any continued operation of the vehicle until the defect is remedied.

These emergency stop-driving orders impose severe costs across all automotive stakeholders:

- **Consumers** face immediate loss of personal transportation, severe logistical disruptions, and life-threatening physical hazards (such as high-speed steering detachment or airbag inflator ruptures).
- **Automakers (OEMs)** incur massive operational and financial penalties, including providing urgent loaner/rental vehicle fleets, expedited parts sourcing, intensive warranty claims, regulatory scrutiny, brand reputational damage, and substantial legal liability.
- **Regulators (NHTSA)** must rapidly assess competing technical filings to prioritize emergency consumer advisories.

Given these asymmetric costs, relying solely on reactive, post-investigation determinations leaves a critical gap in risk management. Developing an automated, predictive methodology to identify which recall filings carry a high risk of escalating to a DND advisory provides manufacturers and regulators with an essential early-warning triage tool.

Core Business & Analytical Questions

1. **Predictive Feasibility:** Can an automated machine learning classification pipeline accurately predict whether an incoming recall filing will warrant a "Do Not Drive" advisory using filing narratives, component systems, production volumes, and manufacturer metadata?
2. **Feature Attribution & Risk Drivers:** What specific mechanical failure modes, narrative hazard keywords, and operational scales most strongly correlate with an escalated DND outcome?
3. **Operational Utility & Triage:** Can an imbalance-aware model achieve high precision on rare-event safety data to enable reliable automated triage without overwhelming safety compliance teams with false alarms?

Part 2: Data and Variables

Data Sourcing & Sanitization

The primary data is sourced from historical NHTSA flat recall campaign databases (FLAT_RCL_PRE_2010.txt and FLAT_RCL_POST_2010.txt), comprising an initial extract of 327,036 records across 29 attributes. To ensure valid, modern statistical inference, a three-stage sanitization pipeline was applied prior to modeling:

1. **Domain Scoping:** The dataset was filtered strictly to vehicle recalls (RCLTYPECD == 'V'), removing non-vehicle equipment, tires, and child restraints.
2. **Modern Policy Horizon:** The data was restricted to filings submitted from the year 2000 onward (record_year >= 2000). This removes legacy decades (1960s–1990s) during which formal DND advisories were not systematically tracked, eliminating policy-bias dilution of the target variable.
3. **Narrative Sufficiency:** Filings with missing or truncated defect descriptions (narrative length $\leq$ 10 characters) were excluded to preserve high-signal mechanical vocabulary.

The resulting sanitized dataset contains **252,573 vehicle records** across **18,999 distinct recall campaigns (CAMPNO)**, with a modern baseline DND occurrence rate of **0.8342%**.

Key Modeling Variables

Variable Name	Data Type	Analytical Role	Operational Definition & Preprocessing
target	Binary (0/1)	Target / Outcome	Derived from DO_NOT_DRIVE. Encoded as 1 if DO_NOT_DRIVE == 'Yes', and 0 otherwise. Severe class imbalance (~0.66% in training).
CAMPNO	Categorical	Grouping Key	Unique NHTSA recall campaign identifier. Used as the grouping key for GroupShuffleSplit to prevent multi-row campaign data leakage across data partitions.

text_narrative	Unstructured Text	Predictor	Concatenation of defect description (DESC_DEFECT) and failure consequence (CONEQUENCE_DEFECT). Transformed using TF-IDF (1,000 max n-grams, custom NHTSA administrative stop words, max_df=0.40, min_df=10).
component	Categorical	Predictor	Parsed from COMPNAME to represent the primary vehicle system (e.g., STEERING, SERVICE BRAKES, AIR BAGS). One-hot encoded with min_frequency=50.
MFGNAME	Categorical	Predictor	Cleaned vehicle manufacturer name. One-hot encoded to capture brand cohorts and historical crisis concentration.
INFLUENCED_BY	Categorical	Predictor	Initiating entity (e.g., MFR, ODI, OVSC), capturing voluntary automaker submissions versus agency-influenced investigations.
POTAFF_log	Continuous	Predictor / Modifier	Natural log of potential affected vehicles: $\ln(POTAFF + 1)$. Normalizes extreme skewness and serves as an interaction moderator between small, acute batches and mass fleet updates.

Data Partitioning Strategy (Preventing Data Leakage)

A single NHTSA recall campaign typically encompasses dozens of vehicle make, model, and year combinations, resulting in multiple records sharing identical campaign numbers (CAMPNO) and defect text narratives. Standard row-level random splitting would distribute identical campaign descriptions across training and evaluation sets, leading to severe data leakage and artificially inflated metrics.

To ensure genuine out-of-sample generalization, a **Group-Aware Split** on CAMPNO was executed:

- **Training Set:** 189,615 records (~70% of data), with a DND incidence rate of **0.6582%**.
- **Validation Set:** 29,055 records (~15% of data), with a DND incidence rate of **0.9155%**.
- **Held-Out Test Set:** 33,903 records (~15% of data), with a DND incidence rate of **1.7491%**.